The Physics of Traffic Congestion and Road Pricing in Transportation Planning

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What is Traffic Congestion?

Congestion occurs when Traffic Demand > Roadway Capacity

Congestion results in queueing, slower travel times, (perhaps) reduced throughput, schedule delay, and wasted resources.

Two obvious solutions:

(1) Increase capacity

(2) Reduce demand



What is Capacity?

Capacity is the maximum number of vehicles that can move past a point (bottleneck) in a given time (e.g. veh/hour).

This is determined by the nature of the vehicles and by the size of the bottleneck.

How to increase capacity?

(1) Increase size of bottleneck (constrained by cost of land, construction, topology, etc.)

(2) Better organize vehicles to move with less “friction”. Constrained by abilities of drivers and cars. Intelligent Transportation Systems aim to do this.

Capacity

Is a property of both the road and the drivers/vehicles.

People drive (ideally) to minimize travel time while ensuring safety, i.e. ensure a safe stopping distance, given perception reaction time of individuals (this varies), vehicle braking time (which depends on vehicle and road conditions). Gaps between vehicles (especially large gaps) are lost capacity.

People are not deterministic in the way that rocks are. From here out however, we will assume they are, and that there is a fixed capacity.

Demand

Number of Trips (trip generation)

Destination of Trips (trip distribution models)

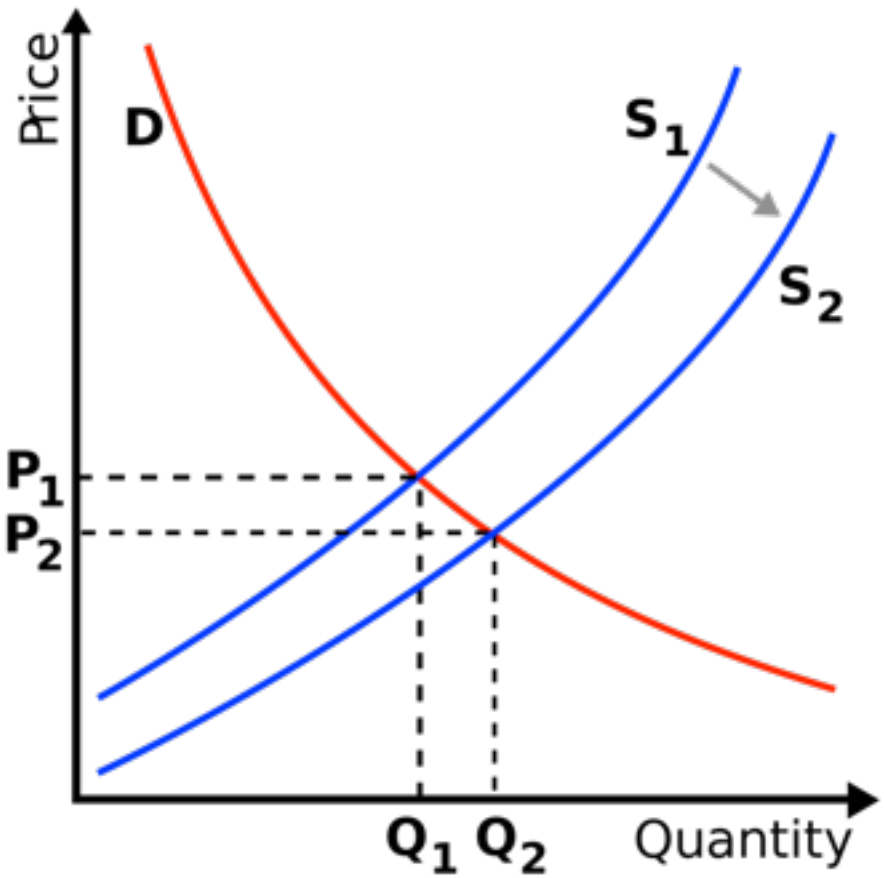
Mode of Trips (mode choice models)

Time of Trips (departure time choice models, bottleneck model)

Route of Trips (network route choice models)

All effected by the time and cost of travel

(Induced Demand)



If you increase capacity, and reduce travel times, you will attract traffic. This must be considered in any real analysis.

Types of Pricing

Coarse Fine	*Average cost*	*Profit-Maximizing**	*Marginal cost*
General with Uniform Links	*Gas Tax*		
Facility-specific	*Public Toll Roads*	*Private Toll Roads (Public Toll*	*Microfoundations Model* *Bottleneck Model*
High Occupancy Toll Lanes		I-394	
Area- Based		London Congestion Charge	
Cordon		*Oslo, Bergen, Trondheim*	*Cordon Pricing Model*
General with Differentiated Links	Mileage Based User Fees		*Ideal*

Note: * *Regulated (price-capped) Profit-Maximizing*

S.O. What?

Wardrop's Principles

User Equilibrium: *The journey times in all routes actually used are equal and less than those which would be experienced by a single vehicle on any unused route.* (like a Nash Equilibrium, developed in parallel)

System Optimal: *At equilibrium the average journey time is minimum.*

System Travel Times:

S.O. < U.E.

With pricing S.O. = U.E.

S.O. What?

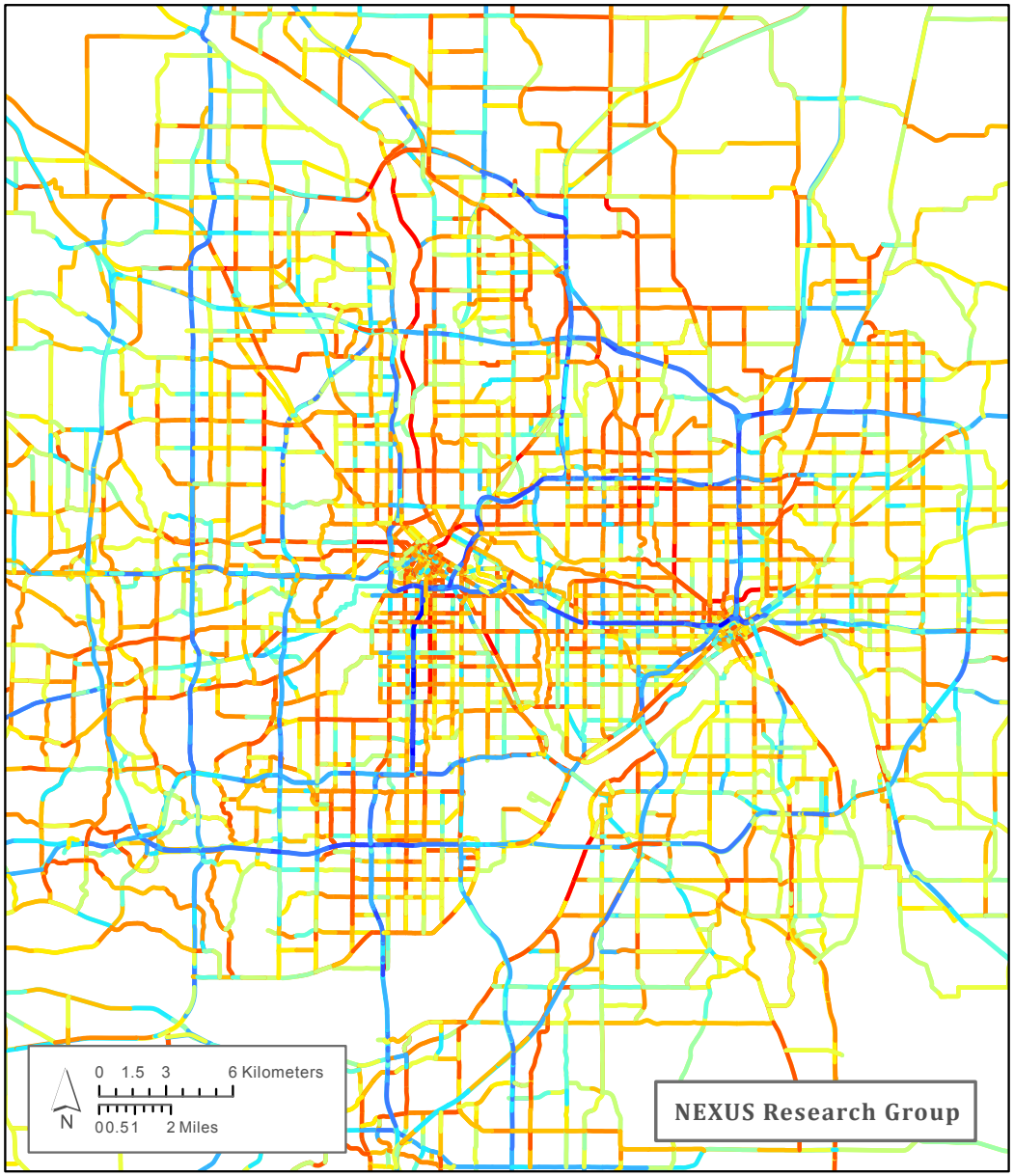
The Price of Anarchy is the ratio of U.E. to S.O. time,

Price of  ≥ 1

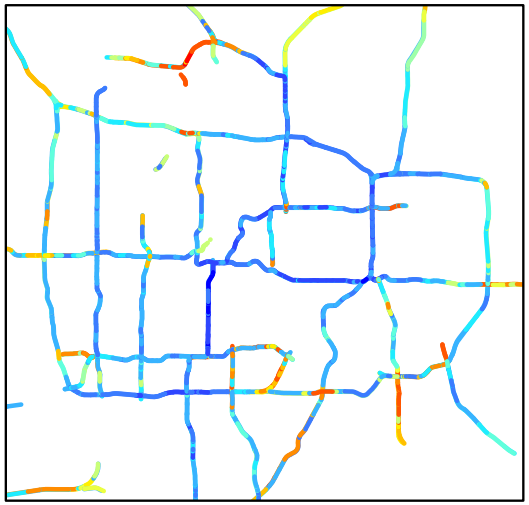
But is it >> 1?

If the Price of Anarchy is small, congestion is more a temporal than spatial phenomenon.

Full Twin Cities Network



Major Freeway Network



Traffic Assignment Differences
7:30 - 8:30 AM
Twin Cities, Minnesota

User Equilibrium Flow Rate
subtracted from
System Optimal Flow Rate

VEHICLE COUNTS

-2414 - -2000	-24 - 0
-1999 - -1000	1 - 25
-999 - -500	26 - 50
-499 - -200	51 - 100
-199 - -100	101 - 200
-99 - -50	201 - 500
-49 - -25	501 - 956

Stats

	User Equilibrium	System Optimal
Vehicle km	9326447	9373185
Vehicle Hours	150964	148389
Speed (km h^{-1})	61.8	63.2
Trips	632232	
Time Savings	1.7%	

Objective of Research

- To build simplest model that explains congestion phenomenon and shows implications of congestion pricing.
- Use game theory to illustrate ideas, informed by structure of congestion problems
  - simultaneous arrival;
  - arrival rate > service flow;
  - first-in, first-out queueing,
  - delay cost,
  - schedule delay cost

Two boats, One lock



Game Theory Assumptions

Actors are instrumentally rational

(actors express preferences and act to satisfy them)

Common knowledge of rationality

(each actor knows each other actor is instrumentally rational, and so on)

Consistent alignment of beliefs

(each actor, given same information and circumstances, would make same choice)

Actors have perfect knowledge

Two-Player Congestion Game

Penalty for Early Arrival (E), Late Arrival (L), Delayed (D)

Each vehicle has option of departing (from home) early (e), departing on-time (o), or departing (l)

If two vehicles depart from home at the same time, they will arrive at the queue at the same time and there will be congestion. One vehicle will depart the queue (arrive at work) in that time slot, one vehicle will depart the queue in the next time slot.

Congesting Strategies

- If both individuals depart early (a strategy pair we denote as *ee*), one will arrive early and one will be delayed but arrive on-time. We can say that each individual has a 50% chance of being early *or* being delayed.
- If both individuals depart on-time (strategy *oo*), one will arrive on-time and one will be delayed and arrive late. Each individual has a 50% chance of being delayed *and* being late.
- If both individuals depart late (strategy *ll*), one will arrive late and one will be delayed and arrive very late. Each individual has a 50% change of being delayed *and* being very late.

Payoff Matrix

		Vehicle 2		
		Early	On-time	Late
Vehicle 1	Early	[0.5*(E+D), 0.5*(E+D)]	[E, 0]	[E ,L]
	On-time	[0, E]	[0.5*(L+D), 0.5*(L+D)]	[0, L]
	Late	[L, E]	[L, 0]	[L+0.5*(L+D), L+0.5*(L+D)]

Note: [Payout for Vehicle 1, Payout for Vehicle 2]

Objective to Minimize Own Payout, S.t. others doing same

Example: (3,1,4)

E, D, L
3, 1, 4

		Vehicle 2		
		Early	On-time	Late
Vehicle 1	Early	[2,2]	*[3,0]*	[3,4]
	On-time	*[0,3]*	[2.5, 2.5] *	[0,4]
	Late	[4,3]	[4,0]	[6.5, 6.5]

Note: * Indicates Nash Equilibrium

Italics indicates social welfare maximizing solution

Payoff matrix with congestion pricing

		Vehicle 2		
		Early	On-time	Late
Vehicle 1	Early	[0.5*(E+D)+τ_e, 0.5*(E+D) + τ_e]	[E, 0]	[E, L]
	On-time	[0, E]	[0.5*(L+D)+τ_o, 0.5*(L+D) + τ_o]	[0, L]
	Late	[L, E]	[L, 0]	[L+0.5*(L+D)+τ_l, L+0.5*(L+D) + τ_l]

What are the proper prices?

- Normally use marginal cost pricing
  - $MC = \partial TC / \partial Q$
- But Total Costs (TC) are discrete, so we use incremental cost pricing
  - IC = $\Delta TC / \Delta Q$
- Total Costs include both delay costs as well as schedule delay costs.
  - $\tau_o = \tau_l = 0.5*(L+D)$
  - $\tau_e = 0.5*(D)$.

Subtleties

- Vehicles may affect other vehicles by causing them to change behavior.
- Total costs do not include these “pecuniary” externalities such as displacement in time, just what the cost would be for that choice, given the other person is there, compared with the cost for that choice if one player were not there.
- You can’t blame departing early on the other player.

Example (3,1,4) with congestion prices

E, D, L

3,1,4

Vehicle 1

	Vehicle 2: Early	Vehicle 2: On-time	Vehicle 2: Late
Early	[2.5,2.5]	[3,0]*	[3,4]
On-time	[0,3]*	[5,5]	[0,4]
Late	[4,3]	[4,0]	[9,9]

Equilibrium Conditions

State	No Pricing	With Pricing
ee	$E=0$ and $D=0$	$E=0$ and $D=0$
eo, oe	$E \le 0.5*(L+D)$ and $E \le L$	$E \le L$
el, le	$E=0$ and $L=0$	$E=0$ and $L=0$
oo	$D \le L$ and $0.5*(L+D) \le E$	$L \le E$ and $D=0$
ol, lo	$L \le E$ and $L \le D$	$L \le E$
ll	$L=0$ and $D=0$	$L=0$ and $D=0$

Note: Assume penalties E, L, D ≥ 0

Summary

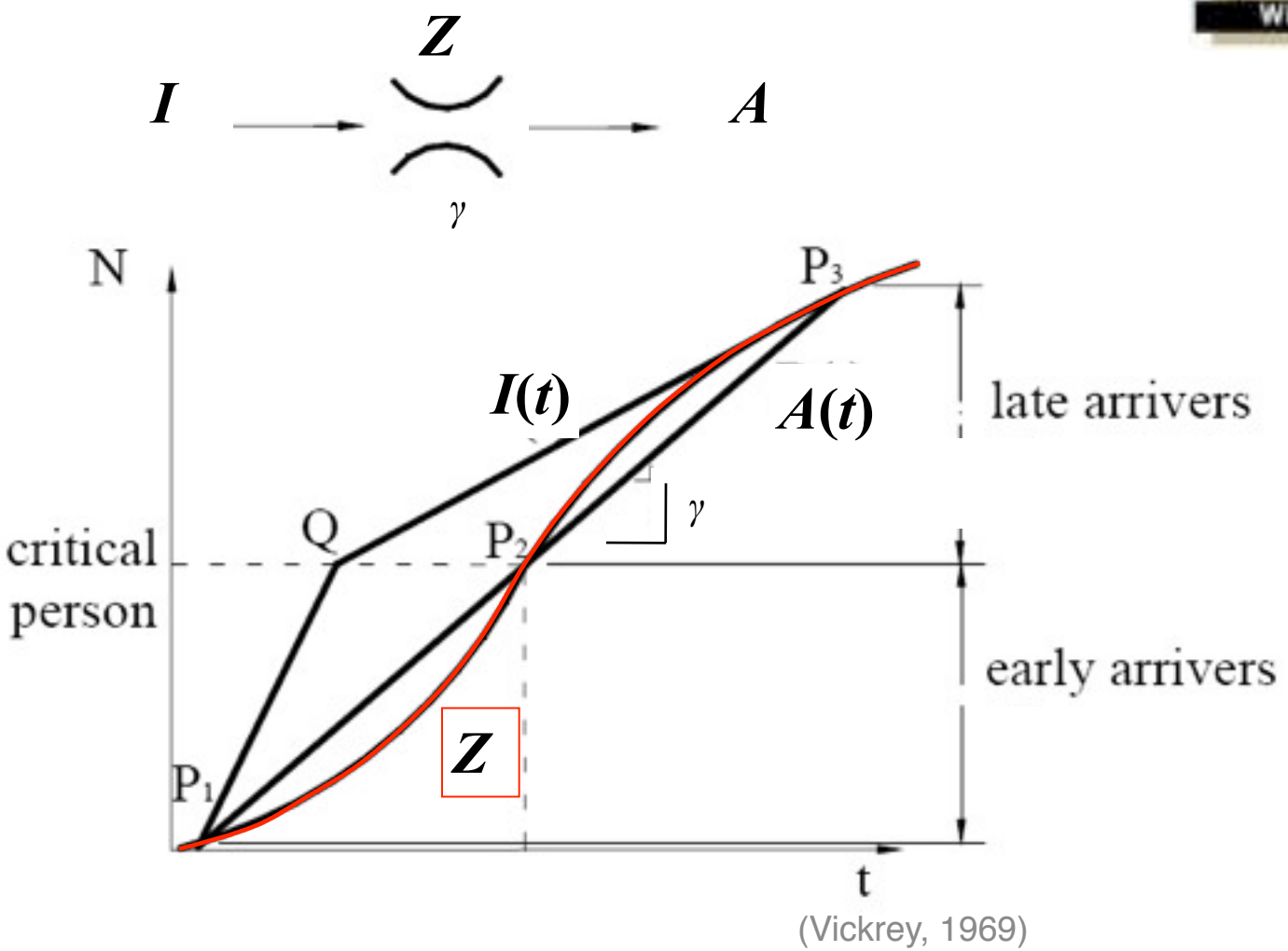
- Presented a simple (the simplest?) model of congestion and pricing.
- A new way of viewing congestion and pricing in the context of game theory.
- Illustrates the effectiveness of moving equilibria from individually to socially optimal solutions.
- Extensions: empirical estimates of E, D, L; risk; uncertainty and stochastic behavior; simulations with more players.

Motivation

- Network-based marginal cost models are not consistent with the physics of traffic
- Bottleneck models cannot describe the overcrowding of networks
- At what mobility level a city should operate?

Motivation

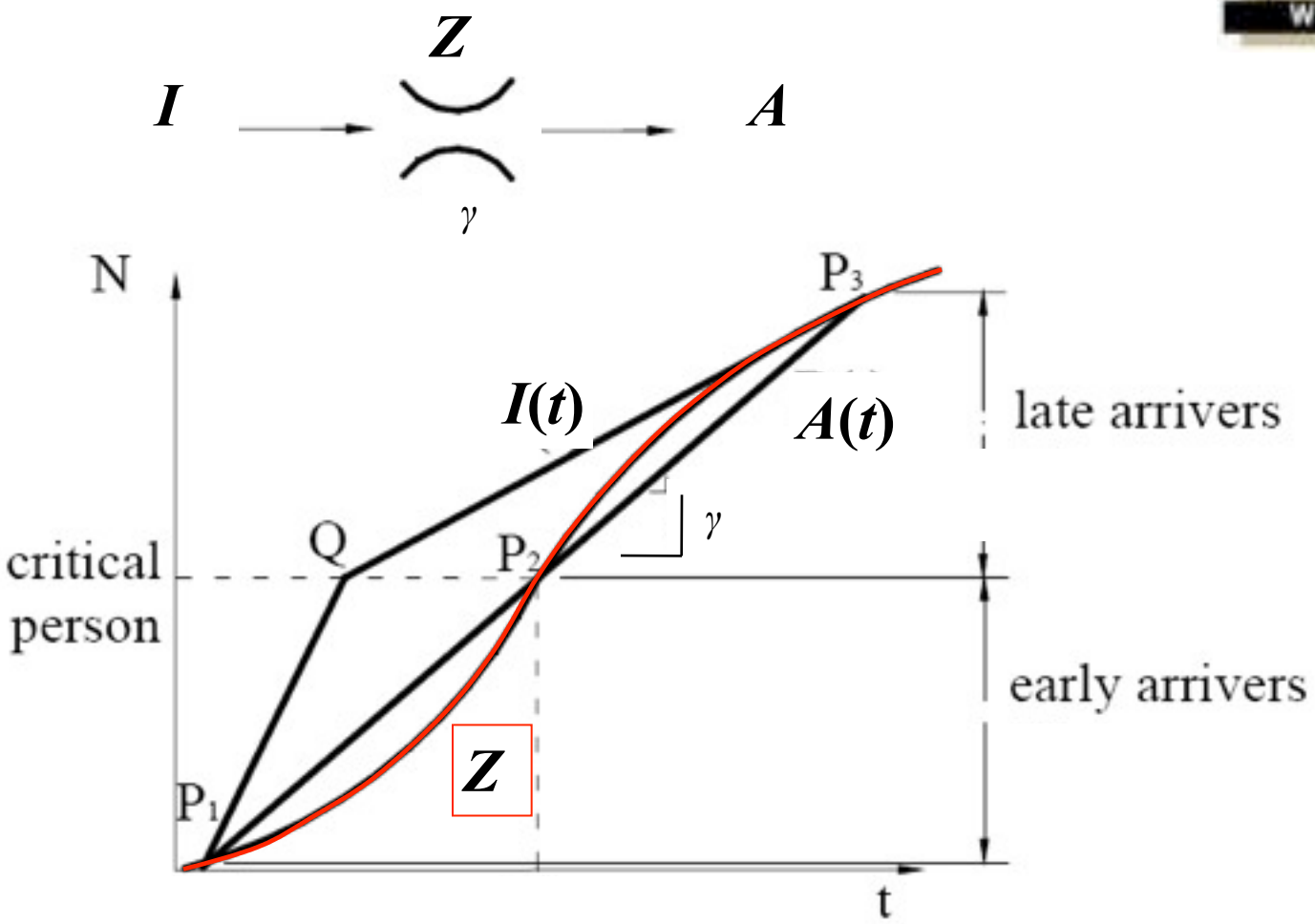




(Vickrey, 1969)

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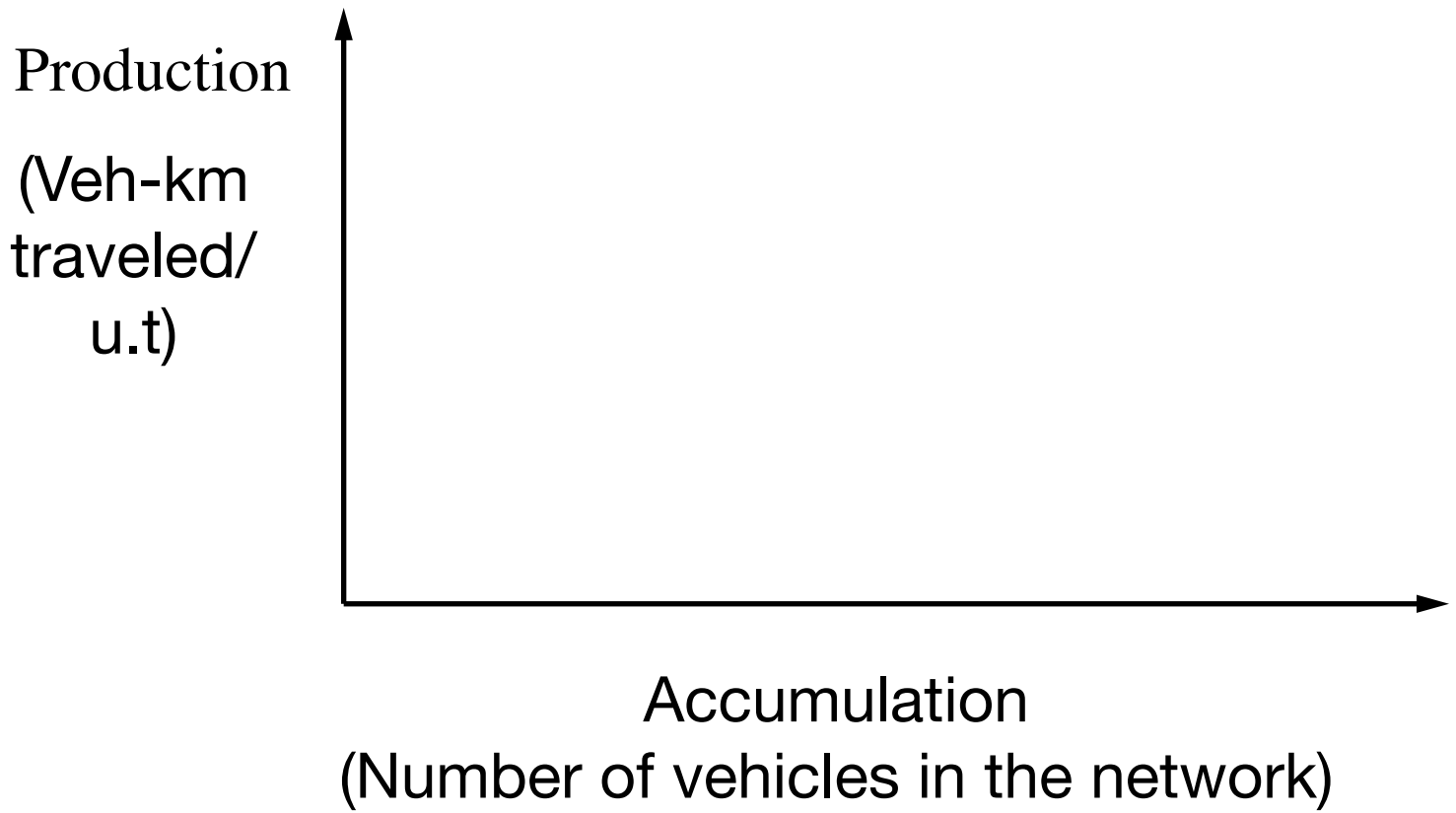




(Vickrey, 1969)

Vickrey's Bottleneck model as above holds for links, BUT: Cities' output decreases with level of congestion.

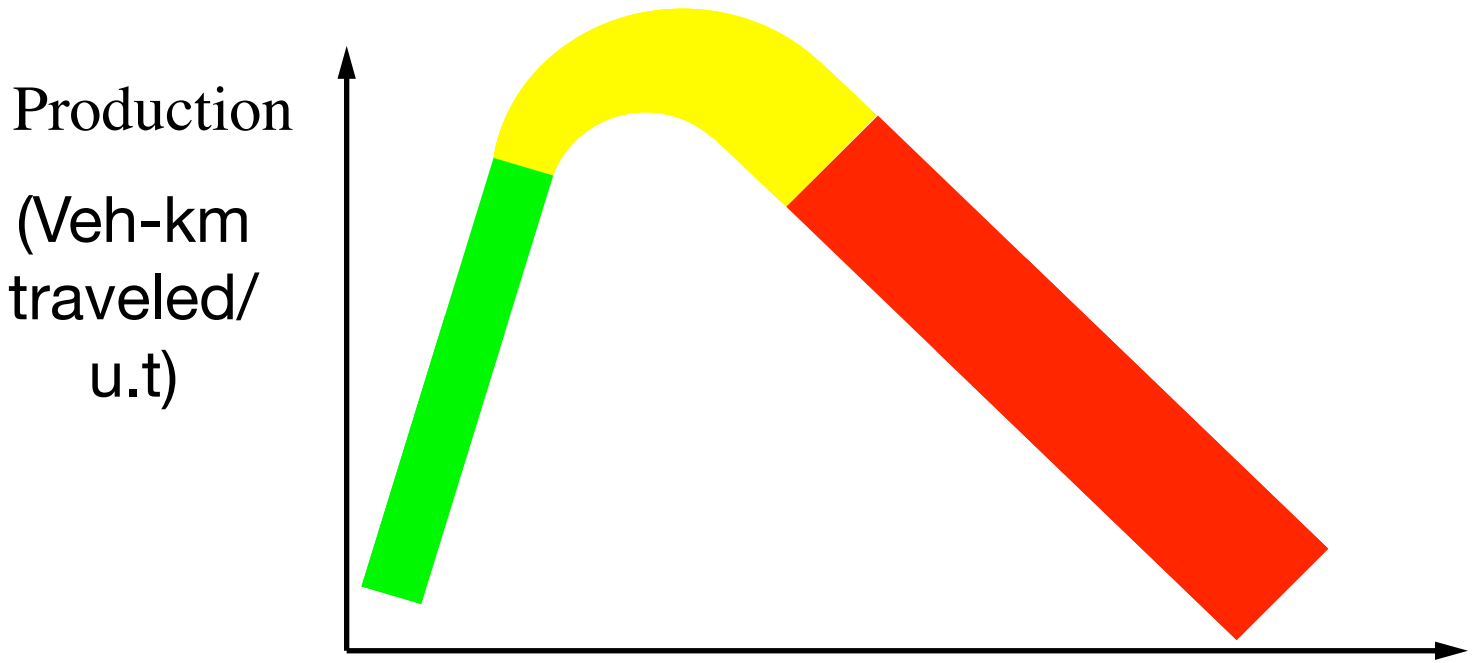
Recent findings



- A Macroscopic Fundamental Diagram (MFD) exists
- Trip completions / Network flow ≃ Constant

(Geroliminis & Daganzo,2007, 2008)

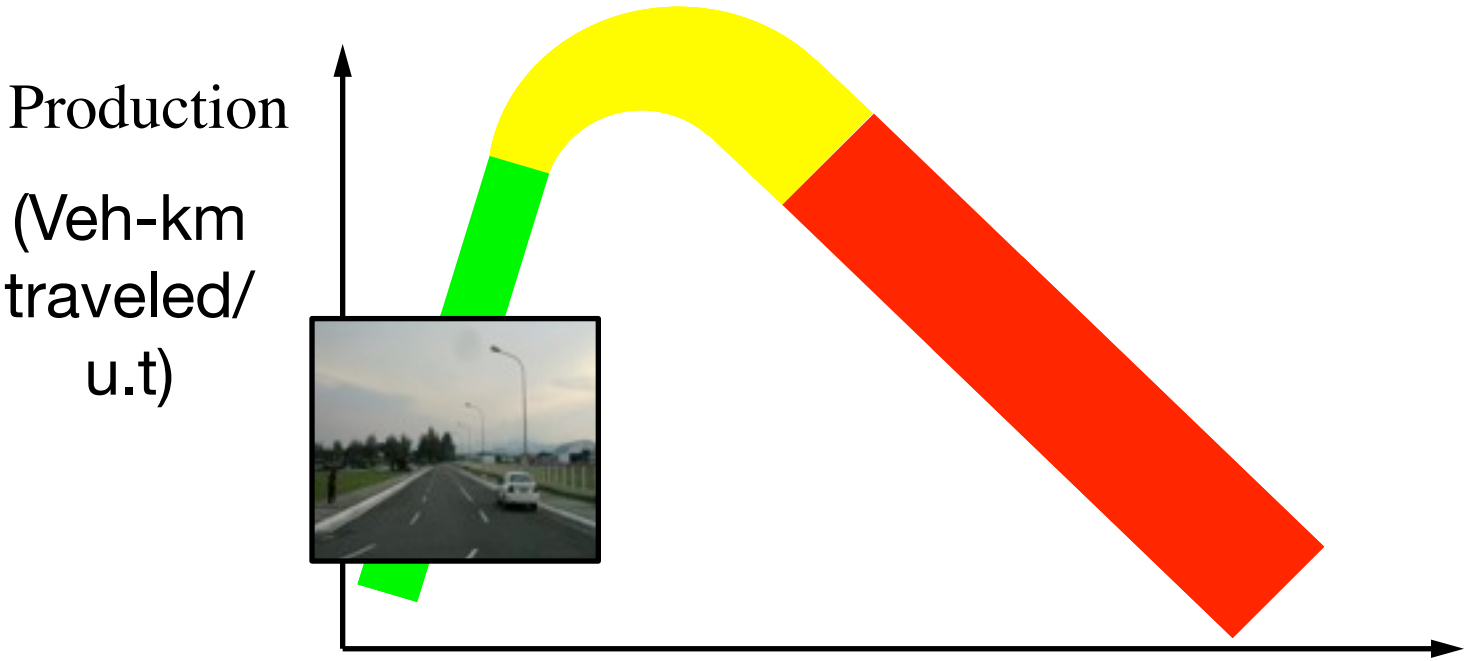
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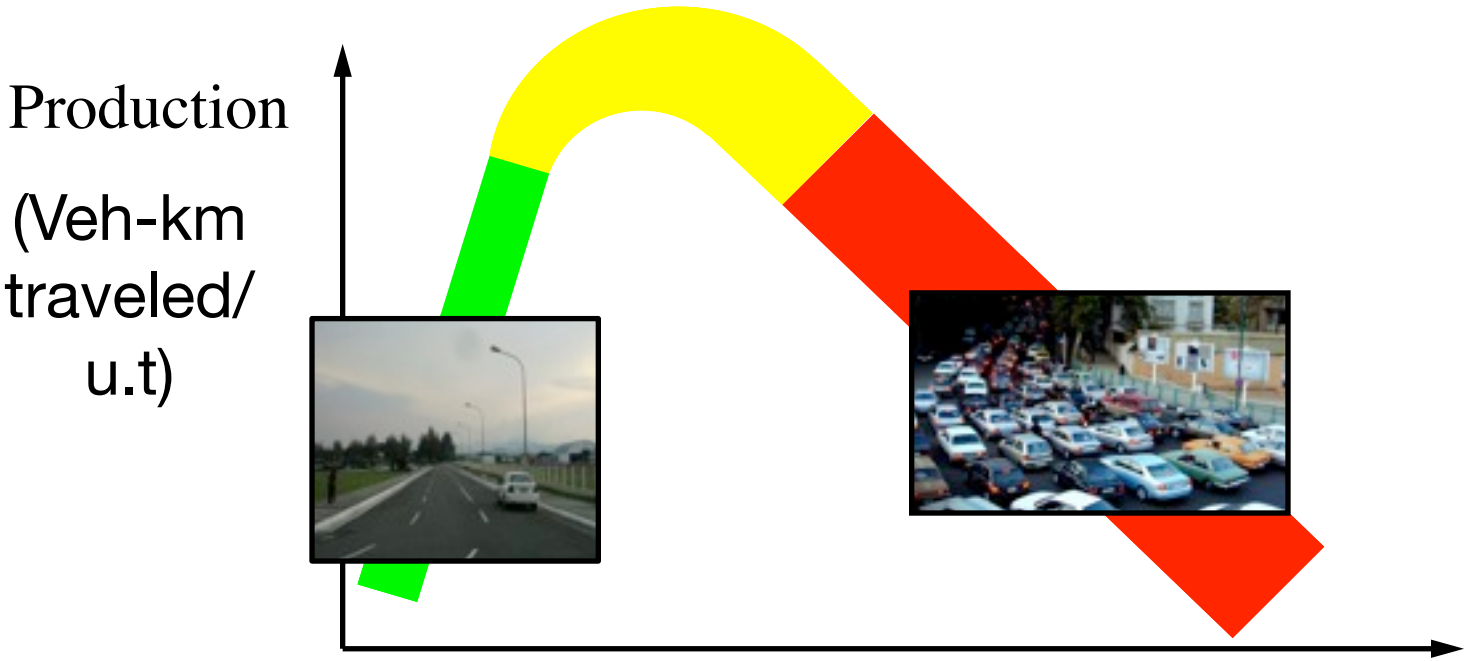
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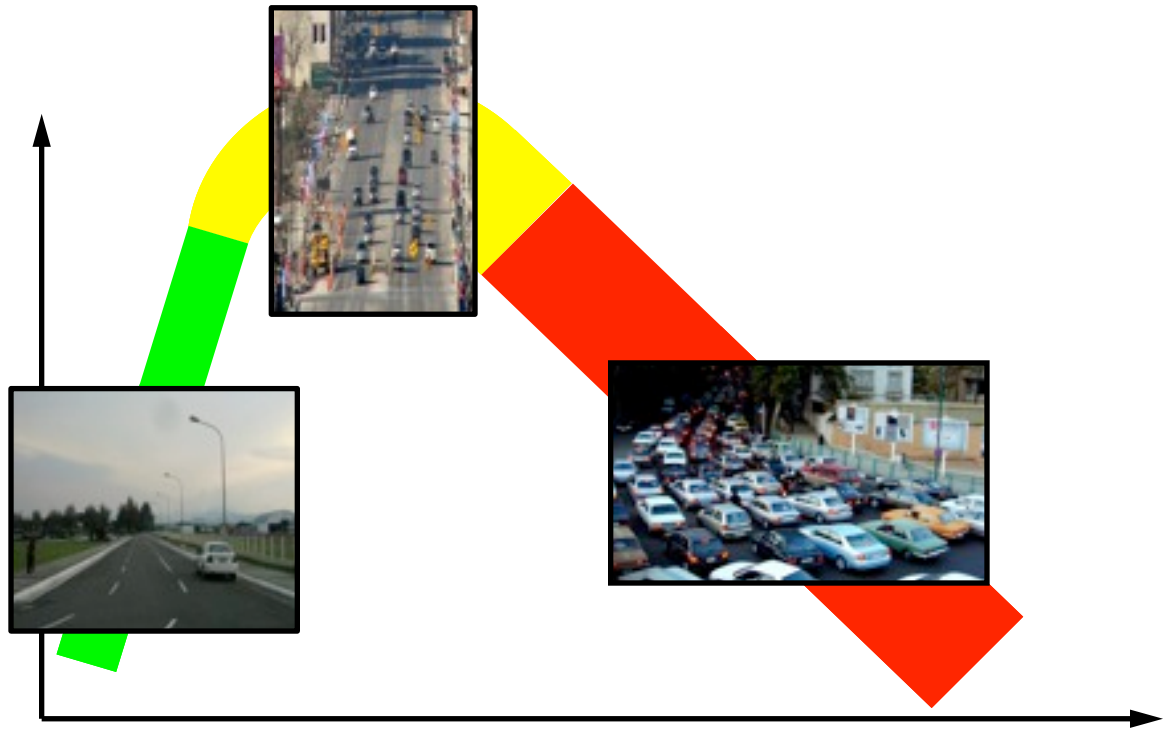
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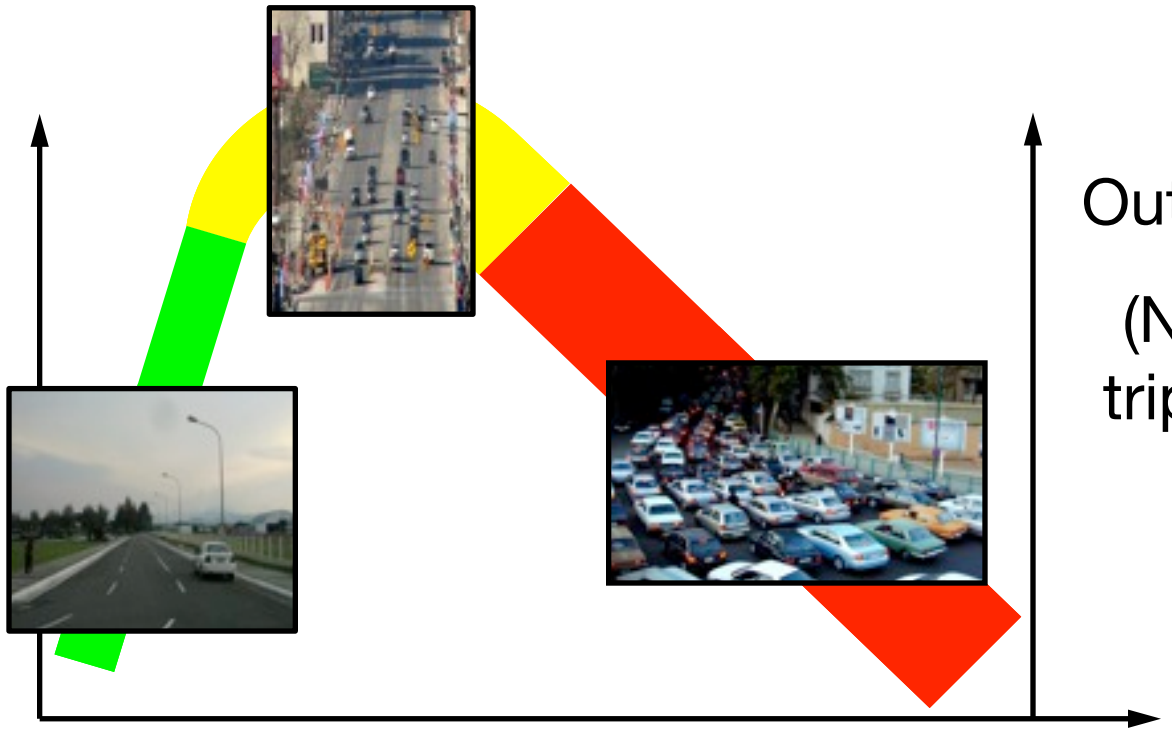
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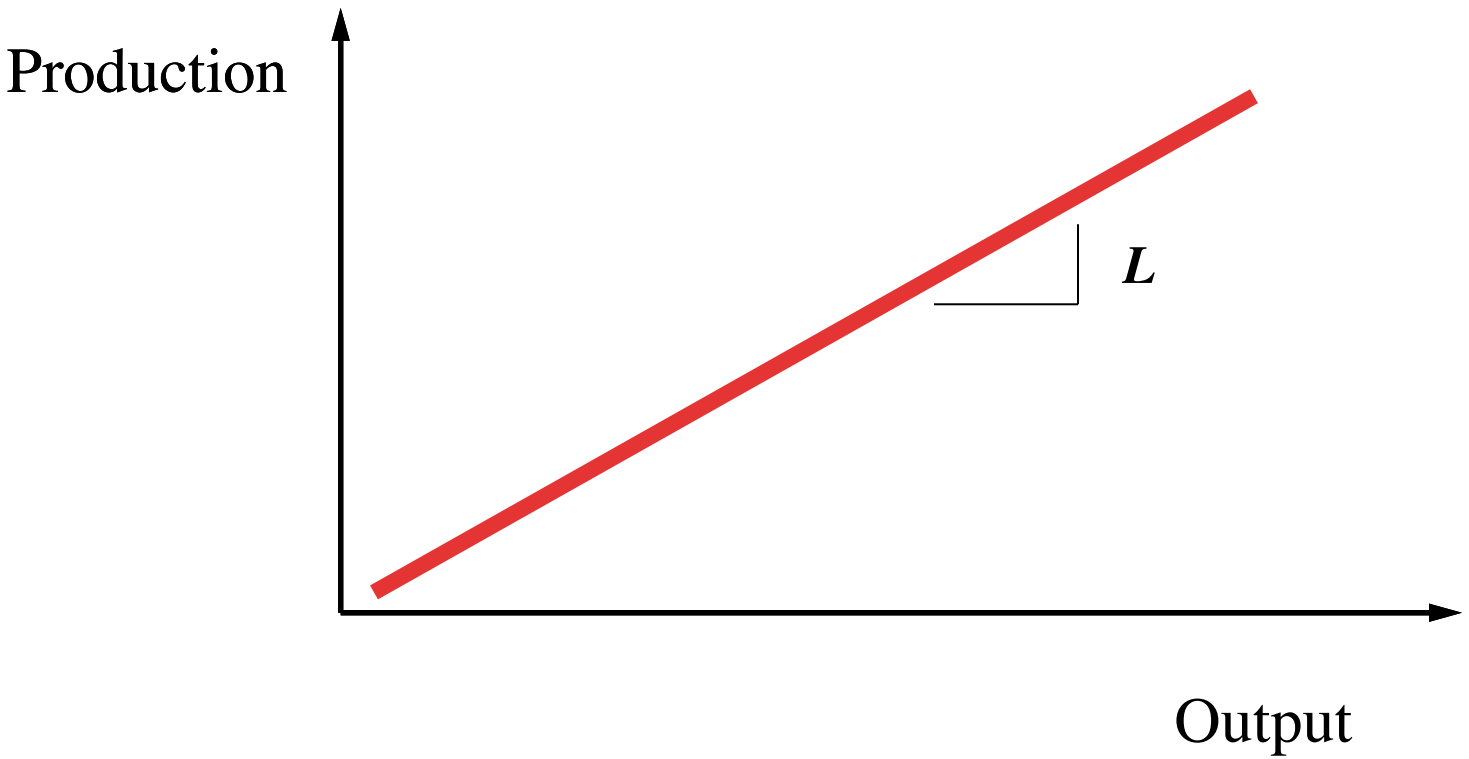
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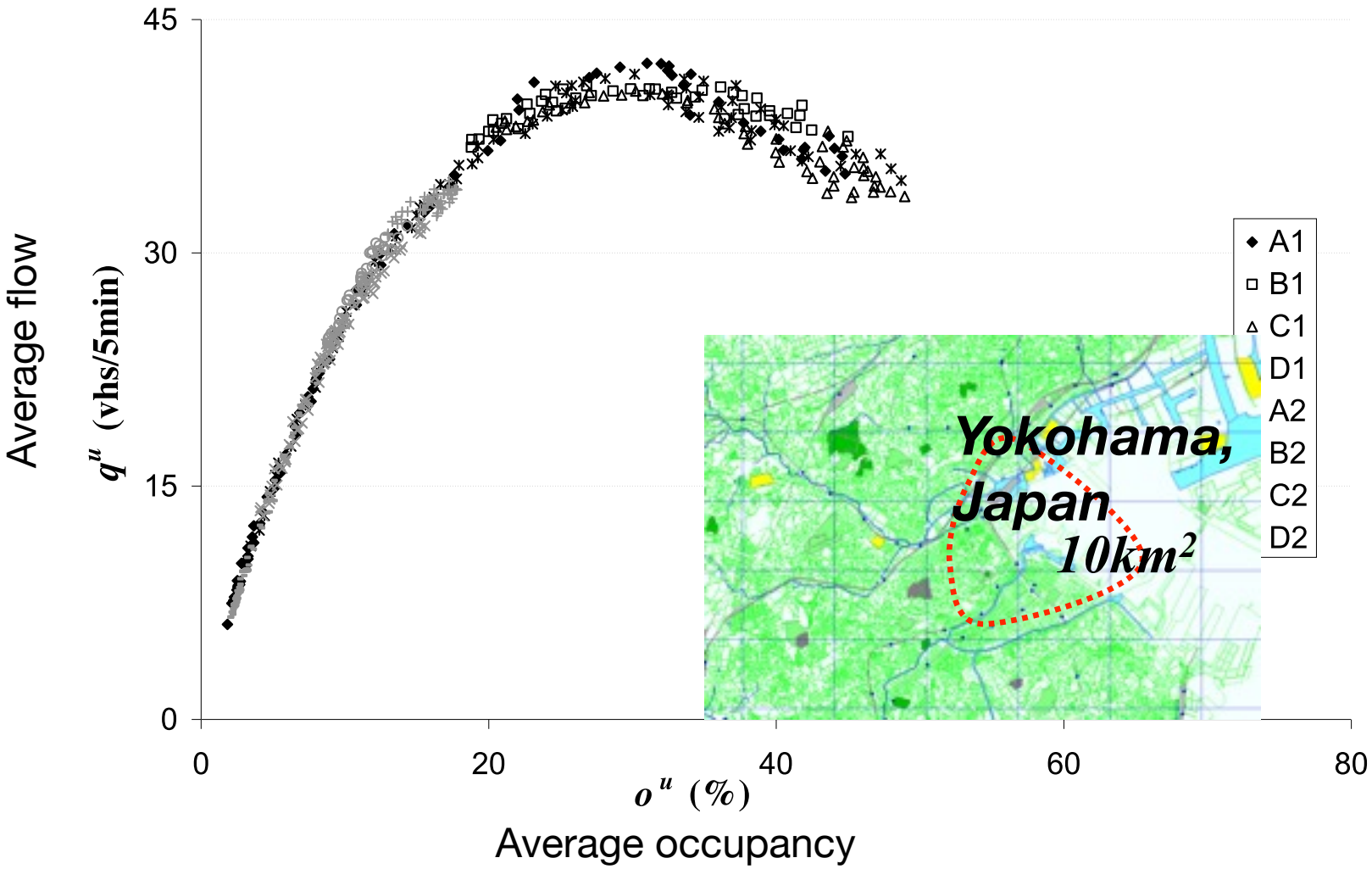
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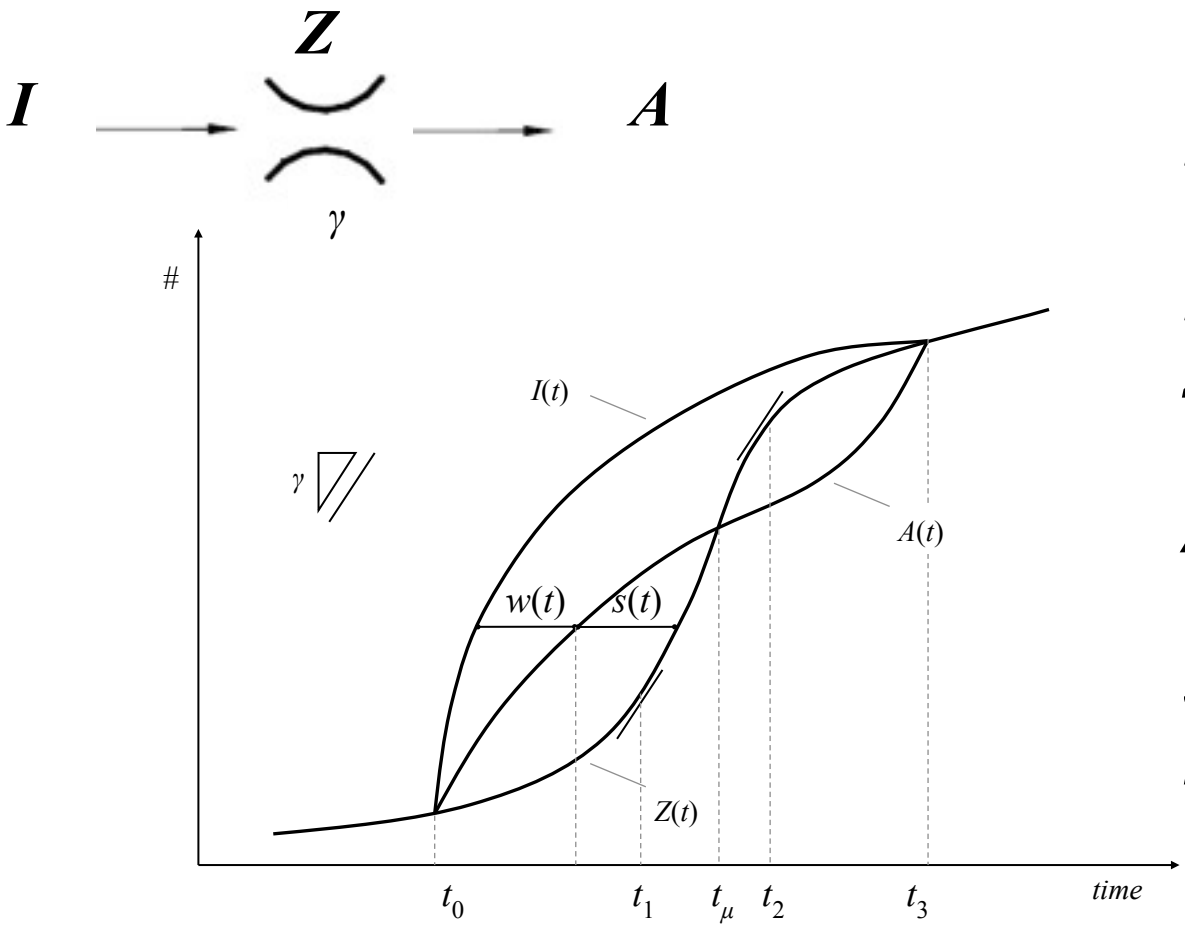
(Geroliminis & Daganzo,2007, 2008)

MFD (Empirical data)



(Geroliminis & Daganzo, 2008)

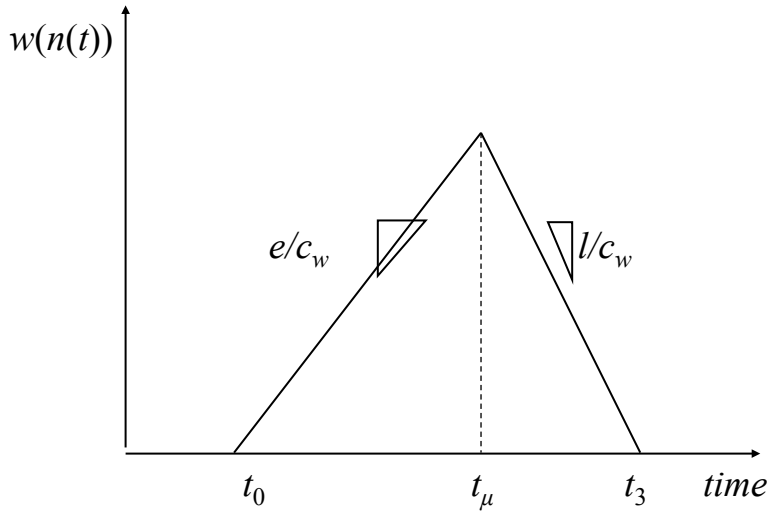
Network equilibrium (No-toll case)



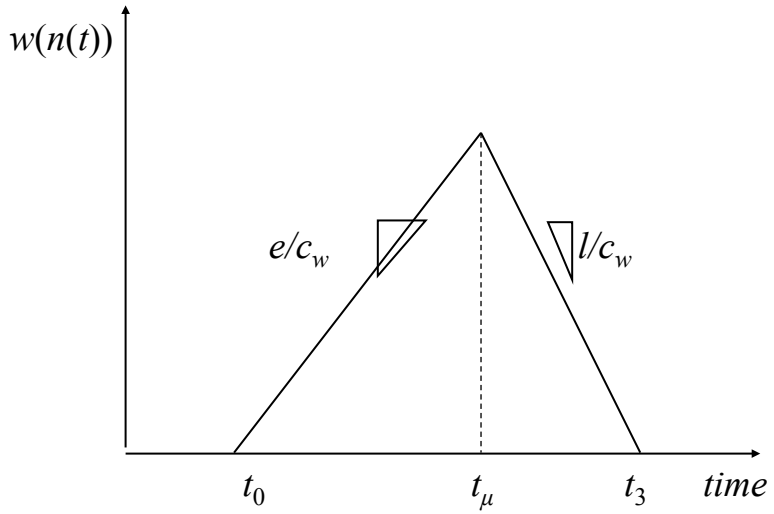
Notation

***I*(t)**: Inflow
***Z*(t)**: Desired arrival rate
***A*(t)**: Arrival rate
w : travel delay
***s*(t)**: schedule delay
n_0: critical accumulation
τ : trip time

Equilibrium Solution



Equilibrium Solution

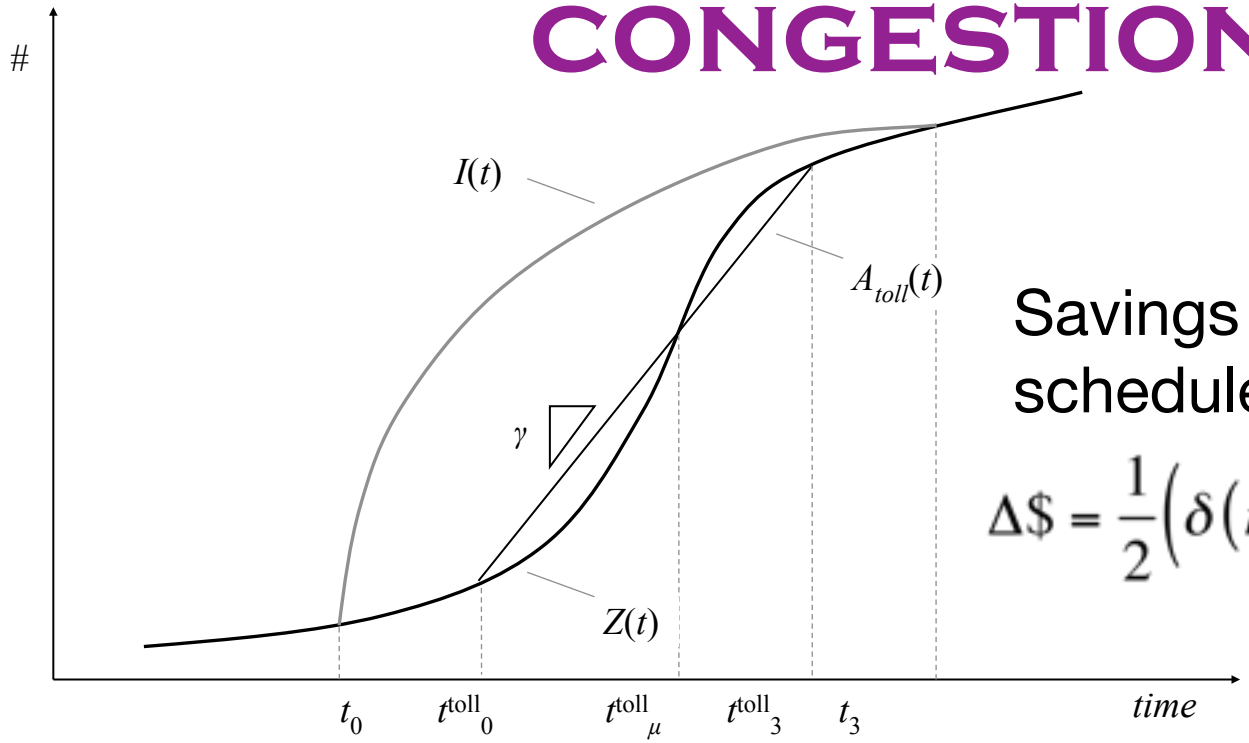


BUT: Demand Congestion interval $[t_1, t_2]$ is a subset of Actual Traffic Congestion interval $[t_0, t_3]$

A constructive solution of t_0, t_3 is described in the paper

t.

Fine toll to reduce congestion

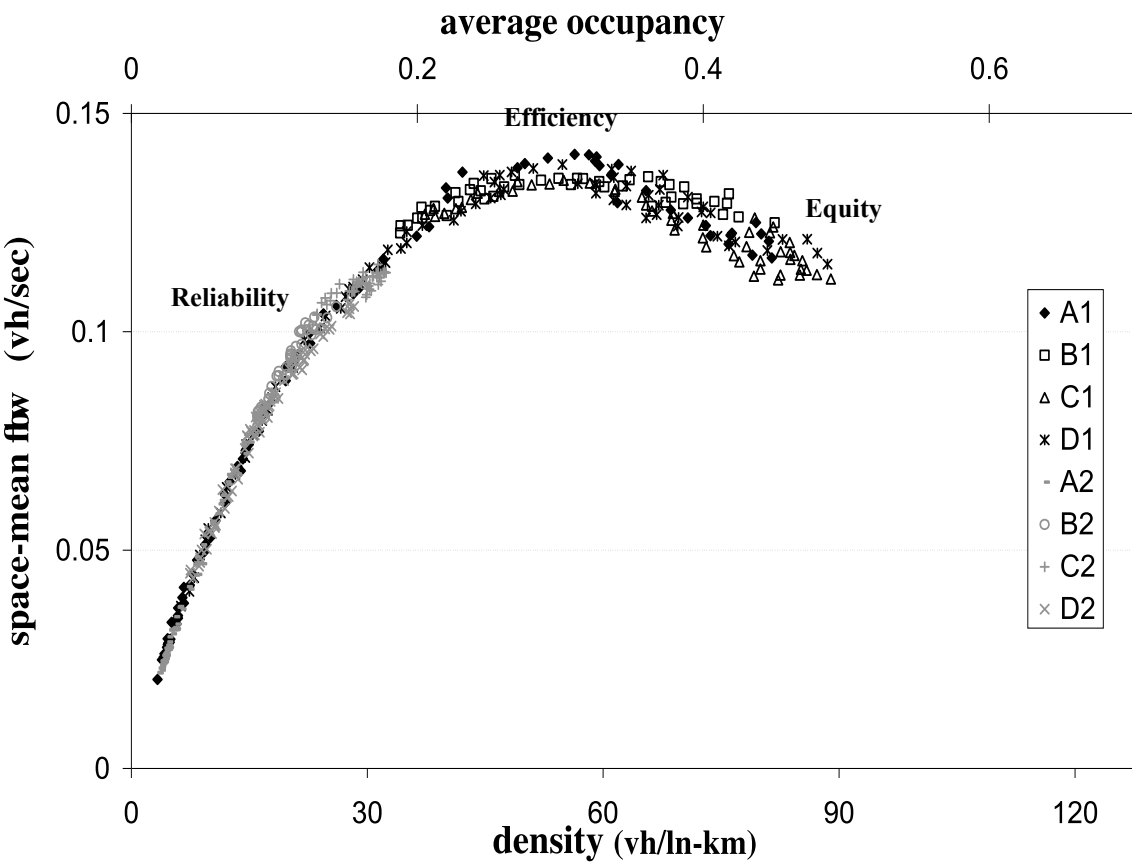


Savings (excluding schedule delay)

$$\Delta\$ = \frac{1}{2}\left(\delta(notoll)^2 - \delta(\gamma)^2\right)\frac{\gamma l e}{l+e}$$

$$T(t) = \begin{cases} \frac{e}{c_w}\left(t - t_0^{toll}\right) & if \quad t_0^{toll} \le t \le t_\mu^{toll} \\ -\frac{l}{c_w}\left(t - t_3^{toll}\right) & if \quad t_\mu^{toll} \le t \le t_3^{toll} \\ 0 & all\ other\ times \end{cases}$$

Policy implications: Equity vs. Reliability



Conclusions

- Constructed parsimonious models of congestion on links, networks.
- Pricing reduces or eliminates congestion, maximizing throughput on networks given interdependence of supply and demand.
- Cordon tolls can be applied to real networks to reduce or eliminate deadweight loss of queueing, prioritize high value trips at times of scarce road capacity. Examples London, Singapore, Stockholm, Oslo.

Papers

- Levinson, David (2005) Micro-foundations of Congestion and Pricing: A Game Theory Perspective.. *Transportation Research part A* Volume 39, Issues 7-9 , August-November 2005, Pages 691-704.
- Zou, Xi and David Levinson (2006) A Multi-Agent Congestion and Pricing Model. *Transportmetrica* Vol.2, No.3, 2006 pp.237-249.
- Zhang, Lei, David Levinson, and Shanjiang Zhu (2008) Agent-Based Model of Price Competition and Product Differentiation on Congested Networks. *Journal of Transport Economics and Policy* Sept. 2008 42(3) pp. 435-461.
- Geroliminis, Nikolas and David Levinson (2009) Cordon Pricing Consistent with the Physics of Overcrowding. Proceedings of Eighteenth International Symposium on Transportation and Traffic Theory, Hong Kong, July 2009.

Thank You

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http://nexus.umn.edu